

A
SEMINAR REPORT
ON
VISUAL POSITIONING SYSTEM

Submitted by

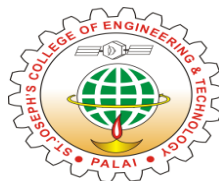
Carolyn Thomas

Reg No. : SJC15CS027

to

the APJ ABDUL KALAM Technological University
in partial fulfillment of the requirements for the award of the degree
of
Bachelor of Technology
in

Computer Science and Engineering



Department of Computer Science and Engineering
ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY
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ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY
PALAI
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the report entitled “**Visual Positioning System**” submitted by **Carolyn Thomas, SJC15CS027** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by her under my guidance and supervision.

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Date : 29th November 2018

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Carolyn Thomas

ABSTRACT

A visual positioning system for indoor locations with associated content is provided herein. The system has a map creator and a viewer. The map creator maps the indoor location by acquiring plans thereof, detects paths through the location and associates with the paths frames relating to objects and views of the paths. The viewer allows a user to orient in the indoor location by locating the user with respect to a path. The viewer enhances GPS/WIFI/3G data by matching user-captured images with the frames, and then interactively displaying the user data from the mapped paths with respect to user queries. We show that it is possible to find optimal routes among several interconnected paths without relying on an external positioning system. Points of interest in a passive environment are detected from real time first-person-view input, which is then processed to pinpoint the current location, and extended further for navigation. Through this paper we explore the advantages and the challenges faced in implementing such a navigation system in a practical scenario. The solution is based on a natural visual search following human methods of locating.

Index Terms: Visual positioning system, User captured images

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Chapter 1

Literature Survey

- i. 3D rendering for training computer vision recognition/Ser. No. 14/140,288/Google Patents

Rendering systems and methods are provided, which generates from received two-dimensional (2D) object information related to an object and 3D model representations [1], a textured model of the object and 3D model representations, a texture model which is placed in training scenes used to generate picture sets which is further used to train image recognition and object tracking.

- ii. 3D space content visualization system/Ser. No. 13/969,352/Google patents

To create three-dimensional visualizations, the visualization system retrieves two-dimensional content data from a content source, defines at least one layer within the content data [2] and defines an anchor and an element in each layer, where in each element represents an object at a three-dimensional boundary.

- iii. Visual positioning system for automated indoor/outdoor navigation/S Anup, Abinav Goel/ieeexplore.ieee.org/document/8228008

It enables an object/user equipped to be able to navigate through closed environments using an automatically generated Spatial Map Graph (SMG) [3] with the aid of pre-placed visual markers. It is possible to find optimal routes among several interconnected paths with relying on an external positioning system.

- iv. Visual Positioning Systems - An Extension to MoVIPS/Chadly Marouane*, Marco Maier†, Sebastian Feld†, Martin Werner† *Research & Development, Munich, Germany

The Mobile Visual Indoor Positioning System (MoVIPS) is a computer-vision-based positioning system using visual feature points for localizing users within indoor environments [8]. Their features include: Pre-filtering with the help of position and orientation methods. Quantization of feature vectors and rotation invariance of the distance estimation.

Chapter 2

Introduction to Visual Positioning System

2.1 Introduction

The Visual Positioning System is defined as the visual- GPS to locate in indoor/outdoor locations. The system provides a tool to locate assets or specific areas within an indoor environment with 100% accuracy. It is based on a natural visual search following human methods of locating. The human eye can recognize places which it previously has seen. Those images matched in the human brain let the people know where they are and how to reach what they desire. Basically the human eye processes a video feed decomposed in frames. Those frames are stored in our memory and we can access Some of them by remembering some of them, although all are already located in our brain storage. This system adopts human behaviour.

Humans need to be fed initially with the indoor environment information through a video or several images. After the environment information is set, through a new pass in the same environment the frames seen are sent to the system. When the system recognizes any frame on memory, it can tell exactly where we are located. Unlike humans, machines are able to process much faster big amounts of information and are not likely to forget any. While for humans it is really difficult to access and process all the frames stored in the brain and analyze them without forget ting any, for a machine it is not a problem at all, it only requires more computational power. Therefore, the present invention emulates basically the human behaviour, using computer vision to process and analyse video frames for proximate matching and recognition request.

2.2 Components of VPS

The system consists of a map creator and a viewer. The map Creator comprises of:

1. Plan module – It is used to configure to acquire at least one plan which is a architectural plan or an image or a set of images of indoor location.
2. Path definer - It is arranged to receive at least one acquired plan from plan module and derive partially therefore the plurality of edges through the indoor location that is to count

the movement of objects or people with respect to the plan derived. Defining paths will be either manually or automatically by a smart path creator (SPC) algorithm.

3. Frame association – It is associated with at least some of the derived paths, a plurality of corresponding frames retrieved from at least one of a video that is fed and also a picture fed. The frames here are associated to the paths which in relation to objects that are depicted in the frames and located along the corresponding paths. The frames may relate to various objects along the paths such as stores, restrooms, exits etc.
4. Tagging module – It allows a user to tag locations or hotspots along the paths. Tagging can be carried out in advance or during the use of the system.

After the operation of map creator, the map is fully configured and can be edited at any time. The next step is to associate a compass to find out the cardinal position with respect to path being derived, to point North to the map. It also comprises of a map manager and search module. Map manager is used to generate, store and retrieve a plurality of maps of the indoor location. A search module to enable searching map manager.

The viewer in VPS is configured to allow a user to orient in the indoor location using the derived paths by locating the user with respect to at least one respective path using GPS. The viewer comprises of:

1. Locator – This is arranged to locate, the user with respect to the plurality of maps of indoor location by receiving, for a user's communication device, a GPS position/WI-FI network location/3G connection location and at least one of the path-associated frames.
2. Display – To present the user with information from the map that corresponds to present the user's location and in relation to the corresponding derived paths. Once the device is located and is able to display all the information the user may want to find something.
3. User Interface – It is used to answer the user queries regarding the derived paths and the objects there along. If include tags, user interface configures to answer user queries regarding tags.
4. Path Finder – An algorithm task that finds the fastest walkable path between two places say the user position and the target.

After this, the user can locate himself with respect to the map and visual information that has been derived.

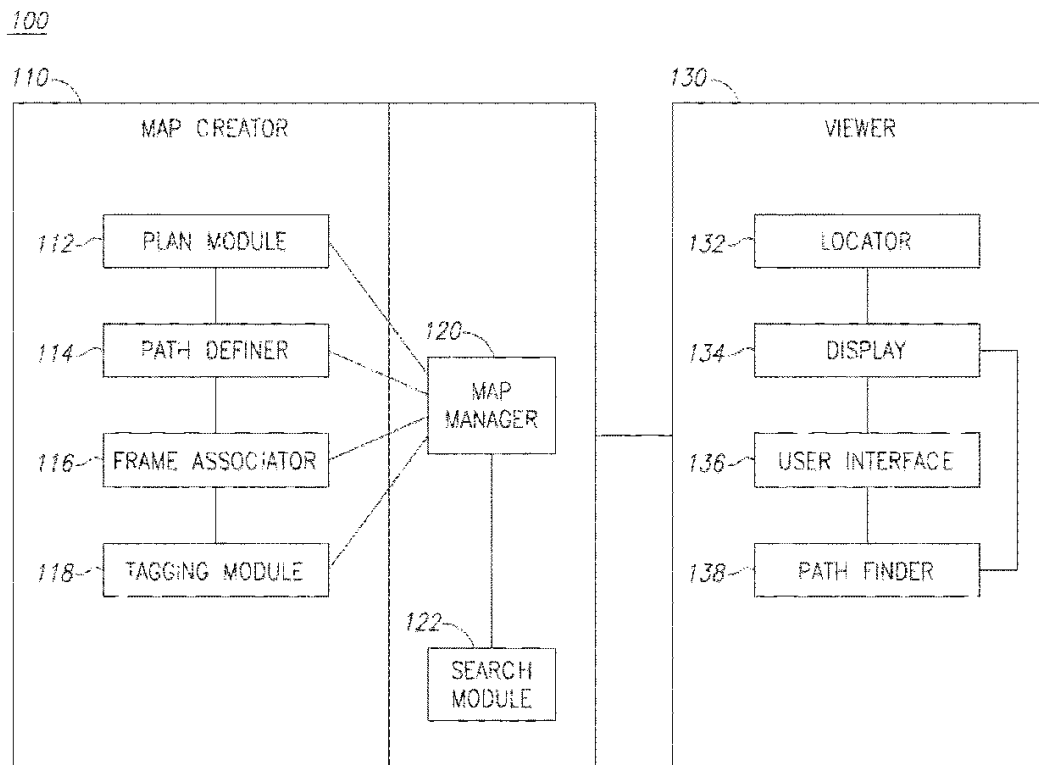


FIG 2A: Schematic Diagram of Visual Positioning System

Chapter 3

Working of VPS

Visual positioning method comprises acquiring at least one plan of an indoor location, deriving at least partially based upon the at least one acquired plan, a plurality of paths through the indoor location and associating with at least some of the derived paths a plurality of corresponding frames retrieved from at least one of a video feed and a picture feed, wherein the frames are associated to paths in relation to objects [9] depicted in the frames and located along the corresponding paths. Visual positioning method comprises providing orientation information to a user relating to the indoor location using the derived paths by locating the user with respect to at least one respective path using GPS/WIFI/3G [8] data enhanced by matching at least one image taken by the user with at least one of the frames and interactively displaying the user data from the mapped paths with respect to user queries. At least part of mapping and providing is carried out by at least one computer processor.

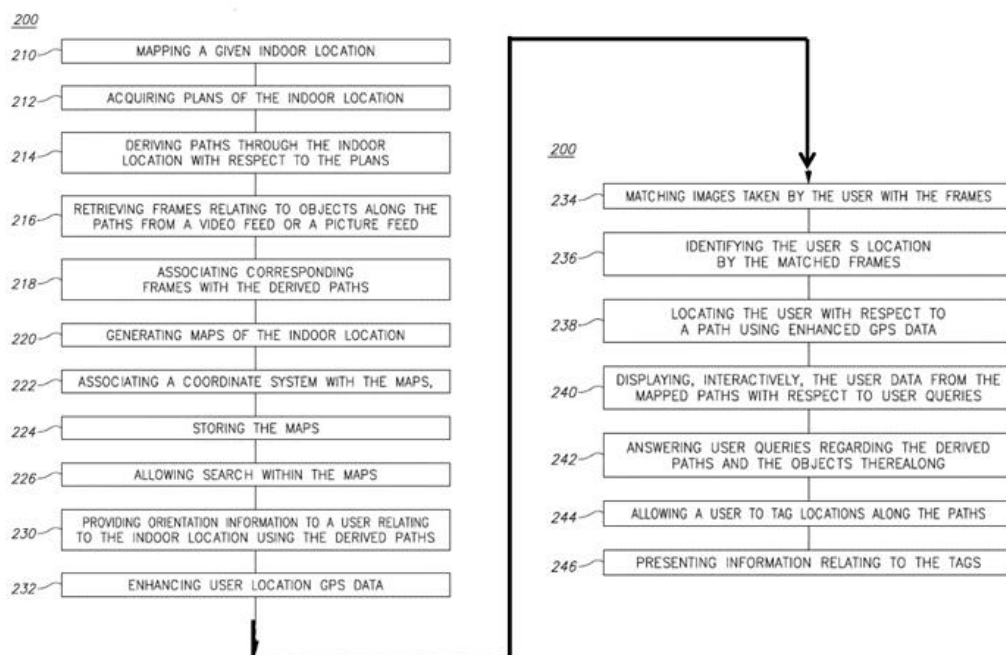


FIG3A: Working of VPS

Let's take an example, consider a user in a book, movies or music store, in a hospital, university, transportation facilities like metro stations, bus stations or airports, cultural places like museums, art centers, music halls, sport places like Stadiums, conferences from all kinds, nursing homes for elderly people, or even corporate buildings of big companies. Basically any user at any indoor place or open small place that the GPS accuracy imperfection can be unsatisfying, or has a location issue, first to know exactly where he is standing in the whole facility and second, how and where to go to reach a specific place, asset, or service he is looking for, so here is a plan of a floor of a building see (FIG 3B).

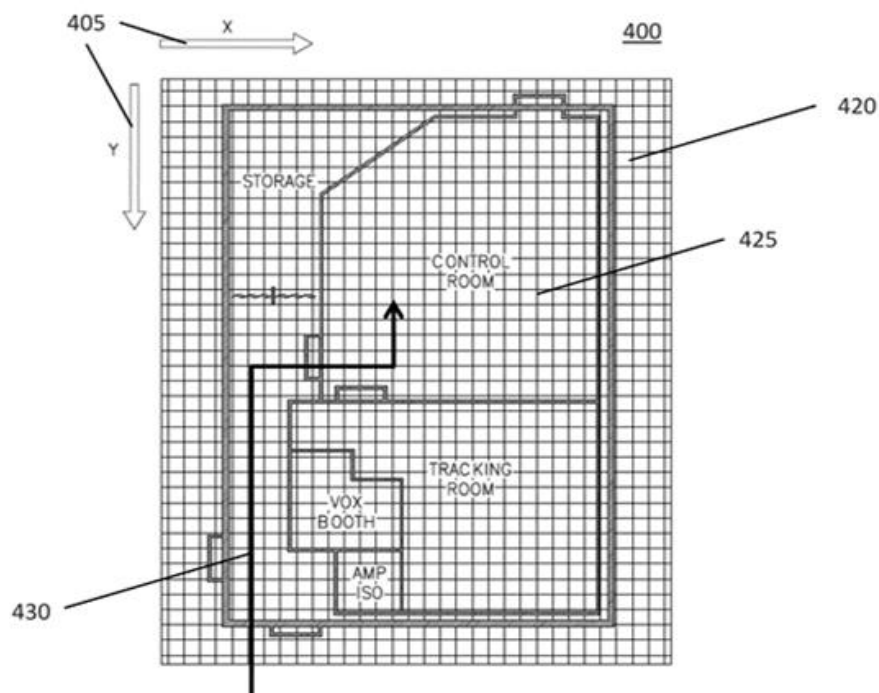


FIG 3B: A floor plan of a building

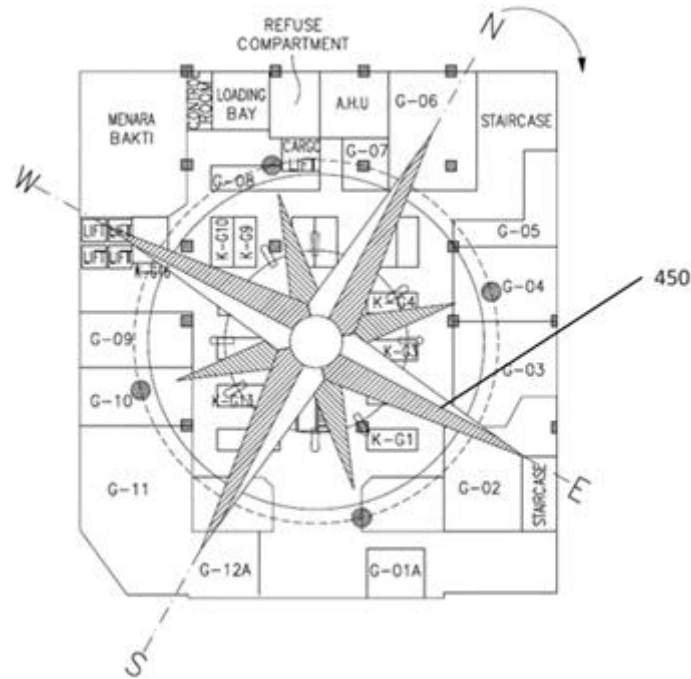


FIG 3C: Compass to find cardinal position

Map creator 110 (Fig 2A) is configured to map a given indoor location by acquiring at least one plan 400 (Fig 3B) of the indoor location; deriving a plurality of paths 430 through the indoor location with respect to at least one plan 400; and associating a plurality of corresponding frames with at least some of derived paths 430. For example, plan 400 may include several blocks 425 such as rooms and passages through which path 430 is selected [9] . Plan 400 may be processed with respect to a grid 405 in which the size of the subunits 420 define the resolution of the paths calculation and image association. Plan 400 may be prepared for each floor of the given indoor location and may be associated with coordinates 405 with respect to the given indoor location. Each plan 400 may be further provided with an interface that relates plan 400 to the cardinal directions 450 (Fig 3C) and allows the user to orient the plan and helps the user to navigate through plan 400.

Chapter 4

VPS in Google Maps

4.1 About present GPS

A global navigation satellite system(GPS) is a constellation of approximately 30 well space satellites that orbit the earth and make it possible for people with ground receivers to pinpoint their geographic location. It consists of three segments

- Satellite constellation
- Ground control network
- User equipment

Baseline satellite constellation consist of 24 satellites. Requires four satellites – 3 for position and 1 for clock respectively. A phone's built-in GPS receiver also communicates with an array of satellites which provides navigation instructions for those either in an automobile or on foot. GPS is used widely by the present generation in use of navigation, say Google Maps (FIG 4A).



FIG 4A: GPS in Google Maps

4.2 Main difference – VPS and GPS

In case of GPS, its accuracy is not assured all the time. That is on the basis of measures: latitude, longitude, altitude it doesn't give an accurate result. But in case of VPS, these three measures are given accurately. So VPS is comparatively more reliable in mobile environment.

4.3 Emerging Application – VPS in Google Maps

Using the concept of VPS and augmented reality(AR), the Google is about to launch an update to Google Maps. They proposed this idea on May,2018. It merged the present interface with live camera view. During navigation, superimposed arrows will appear at each turns making it easier to interpret directions of the respective destination. It also has inserted an animation character to remove doubts as seen in present GPS maps for instance a fox will show the way to your destination (FIG 4B).

Here VPS plays a role by combining our mobile camera with the Google's data to get better sense of position than possible with just a GPS. The augmented reality helps when looking at storefront. This is mainly helpful for the people living in urban areas.

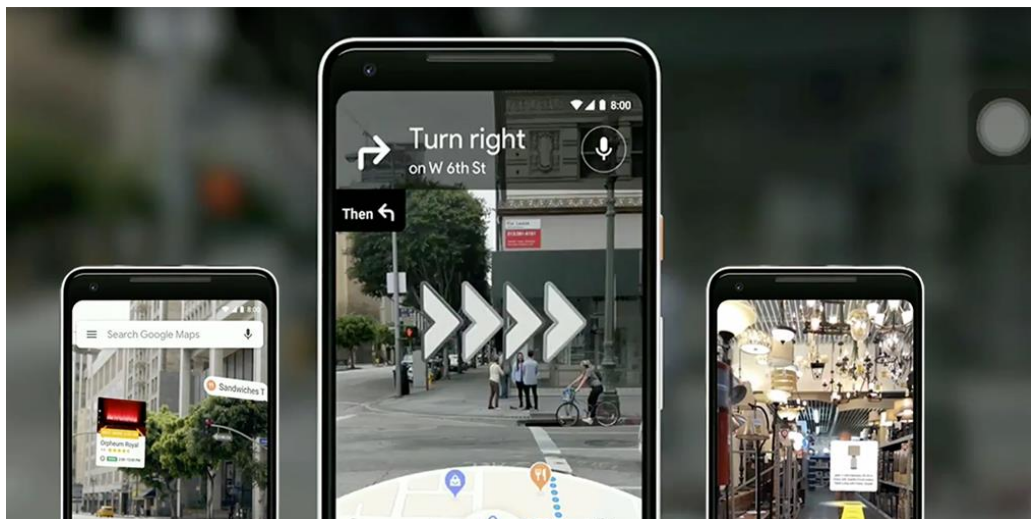


FIG 4B: VPS in Google Map

Chapter 5

Conclusion

This theory of visual positioning system(VPS) can be used in various applications to obtain visually the position of a person in an indoor environment which cannot be provided by the GPS. Visual Positioning can be used in many era's such as in cases of vehicle detection, indoor positioning, mobile searching. A whole system of indoor location including those main features: The system's first part includes a mapping part, a tool to draw a map, drawing the path and assigning the hot spots and tag them; the system has a database of pictures or whole video tagged with places, each picture or set of pictures/frames from video define one place; the system has a recognition engine both in the mobiles and in the server; the server is responsible to recognize where the user is at the beginning; later the part of the database that is close in Some radius around the position of the user, and this way the next recognition is in real time in the mobile; while the user is moving and the system recognizes new spots he is at we download new part of the database and so on; the system can also recognize general objects in the picture/video data base such as stairs/elevators, etc.; and the system draws and tracks positions on the map and may provide directions to any other place in the map. Presently VPS has brought a contribution in updating the google maps which will be to sense positions which was not possible with just GPS. The present system provides a tool to locate assets or specific areas within an indoor environment with 100% accuracy.

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